



Methods in Differential Equations

• Mark Scheme

Core Pure 2 Chapter 7: Methods in Differential Equations. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – First-order equations

1 $\int \frac{1}{y} dy = \int x dx \Rightarrow \ln y = \frac{1}{2}x^2 + c$ M1; $y = Ae^{\frac{1}{2}x^2}$ A1.

2 IF = $e^{\int 2 dx} = e^{2x}$ M1; $\frac{d}{dx}(ye^{2x}) = e^{3x} \Rightarrow ye^{2x} = \frac{1}{3}e^{3x} + c$ M1; $y = \frac{1}{3}e^x + ce^{-2x}$ A1.

3 $\frac{dy}{dx} + \frac{2}{x}y = x$, IF = $e^{\int 2 \ln x} = x^2$ M1; $x^2 y = \int x^3 dx = \frac{1}{4}x^4 + c$ M1; $y = \frac{1}{4}x^2 + \frac{c}{x^2}$ A1.

4 IF = $e^{\int \tan x dx} = e^{\ln \sec x} = \sec x$ M1; $\frac{d}{dx}(y \sec x) = \cos x \sec x = 1$ M1; $y \sec x = x + c$, and $x = 0, y = 2$ gives $c = 2$ A1; $y = (x + 2) \cos x$ A1.

5(a) $\frac{dy}{dx} + \frac{4x}{x^2+1}y = \frac{x}{x^2+1}$ M1; IF = $e^{\int 2 \ln(x^2+1)} = (x^2+1)^2$ M1 A1; $y(x^2+1)^2 = \int x(x^2+1) dx = \frac{1}{4}(x^2+1)^2 + c$ M1; $y = \frac{1}{4} + \frac{c}{(x^2+1)^2}$ A1.

5(b) $0 = \frac{1}{4} + c \Rightarrow c = -\frac{1}{4}$ M1; $y = \frac{1}{4} - \frac{1}{4(x^2+1)^2}$ A1.

5(c) y increases towards $\frac{1}{4}$ (approaches $\frac{1}{4}$ from below) B1.

Section B – Second-order homogeneous equations

6 $m^2 - 5m + 6 = 0 \Rightarrow m = 2, 3$ M1; $y = Ae^{2x} + Be^{3x}$ A1.

7 $4m^2 + 4m + 1 = (2m + 1)^2 = 0$ M1; repeated root $m = -\frac{1}{2}$ A1; $y = (A + Bx)e^{-\frac{1}{2}x}$ A1.

8 $m^2 + 2m + 5 = 0$ M1; $m = -1 \pm 2i$ A1; $y = e^{-x}(A \cos 2x + B \sin 2x)$ A1.

9 $m = \pm 3i$, so $y = A \cos 3x + B \sin 3x$ M1 A1; $A = 2$, and $y' = -3A \sin 3x + 3B \cos 3x$ gives $3B = 6$ M1; $y = 2 \cos 3x + 2 \sin 3x$ A1.

10(a) $m^2 - 2m + 2 = 0$ M1; $m = 1 \pm i$ A1; $y = e^x(A \cos x + B \sin x)$ A1.

10(b) $y(0) = 0 \Rightarrow A = 0$ B1; $y' = Be^x(\sin x + \cos x)$, so $y'(0) = B = 1$ M1; $y = e^x \sin x$ A1.

10(c) $e^x(\sin x + \cos x) = 0 \Rightarrow \tan x = -1$ M1; $x = \frac{3\pi}{4}$ A1.

Section C – Non-homogeneous equations

11 Try $y = \lambda$: $2\lambda = 4$ M1; $y = 2$ A1.

12 Try $y = ax + b$: $a - 2(ax + b) = 4x$ M1; $-2a = 4, a - 2b = 0$ M1; $y = -2x - 1$ A1.

13 CF: $y = A \cos 2x + B \sin 2x$ B1; try $y = \lambda e^x$: $\lambda + 4\lambda = 5, \lambda = 1$ M1; $y = A \cos 2x + B \sin 2x + e^x$ A1.

14 $m^2 - 3m + 2 = 0 \Rightarrow$ CF $y = Ae^x + Be^{2x}$ B1; e^{2x} is in the CF, so try $y = \lambda x e^{2x}$: $y' = \lambda(1 + 2x)e^{2x}, y'' = \lambda(4 + 4x)e^{2x}$ M1; substituting, $\lambda e^{2x}[(4 + 4x) - 3(1 + 2x) + 2x] = \lambda e^{2x} = 3e^{2x}, \lambda = 3$ A1; $y = Ae^x + Be^{2x} + 3xe^{2x}$ A1.

15(a) $m^2 + 2m + 5 = 0 \Rightarrow m = -1 \pm 2i$ M1; CF $y = e^{-x}(A \cos 2x + B \sin 2x)$ A1.

15(b) $y' = -a \sin x + b \cos x, y'' = -a \cos x - b \sin x$; substituting, $(4a + 2b) \cos x + (4b - 2a) \sin x = 10 \sin x$ M1; $4a + 2b = 0, 4b - 2a = 10$ M1; $a = -1, b = 2$ A1.



15(c) $y = e^{-x}(A \cos 2x + B \sin 2x) + 2 \sin x - \cos x$; $y(0) = 0 \Rightarrow A = 1$ $B1$; $y' = e^{-x}[(2B - A) \cos 2x - (2A + B) \sin 2x] + 2 \cos x + \sin x$ $M1$; $y'(0) = 2B - 1 + 2 = 0 \Rightarrow B = -\frac{1}{2}$ $A1$; $y = e^{-x}(\cos 2x - \frac{1}{2} \sin 2x) + 2 \sin x - \cos x$ $A1$.

15(d) $e^{-x} \rightarrow 0$, so $y \approx 2 \sin x - \cos x$: oscillation with constant amplitude $\sqrt{5}$ $B1$.

Roar-level challenge

16 $m^2 - 4m + 4 = (m - 2)^2 = 0$, repeated root $m = 2$ $M1$; CF $y = (A + Bx)e^{2x}$ $A1$; both e^{2x} and xe^{2x} are in the CF, so try $y = \lambda x^2 e^{2x}$ $M1$; $y' = \lambda(2x + 2x^2)e^{2x}$, $y'' = \lambda(2 + 8x + 4x^2)e^{2x}$, substituting gives $2\lambda = 1$, $\lambda = \frac{1}{2}$ $A1$; $y = (A + Bx)e^{2x} + \frac{1}{2}x^2 e^{2x}$ $A1$.